

PATHWISE

Learn it your way!

AI-POWERED PERSONALIZED LEARNING PATHS

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THE PROBLEM

Why One-Size-Fits-All Learning Fails



- **Content Overload Without Direction:**
Millions of tutorials exist, but learners struggle to find the right sequence.



- **Generic & Static Roadmaps:**
Existing learning paths don't account for individual backgrounds, pace, or learning styles.



- **Invisible Skill Gaps:**
Learners don't know what prerequisite fundamentals they are missing.



- **Zero Progress Adaptation:**
When learners struggle or advance fast, static roadmaps never adapt.



- **High Drop-out Rates:**
Learners abandon goals not from lack of interest, but because they are taught the wrong way.

“Every learner has a unique pace, style, and goal. Why should their curriculum be universal?”



OUR SOLUTION

Pathwise is an **AI-powered** platform that creates, adapts, and validates personalized learning paths using **learning science (VARK)** and a **deterministic catalog engine**—ensuring **100% real, relevant courses**.

1 Intelligent Onboarding



- Job
- Internship
- Career Switch
- Skill
- Research
- ... and more

- ✓ 8 goal objectives including Job, Internship, Career Switch, Skill, Research, and more.
- ✓ Custom write-in support for personalized goals.

2 20-Question VARK Assessment



Visual



Auditory



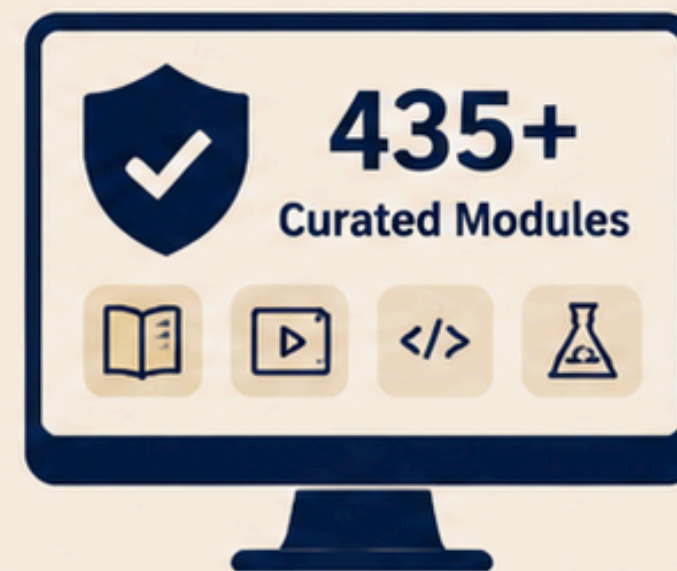
Read/Write



Kinesthetic

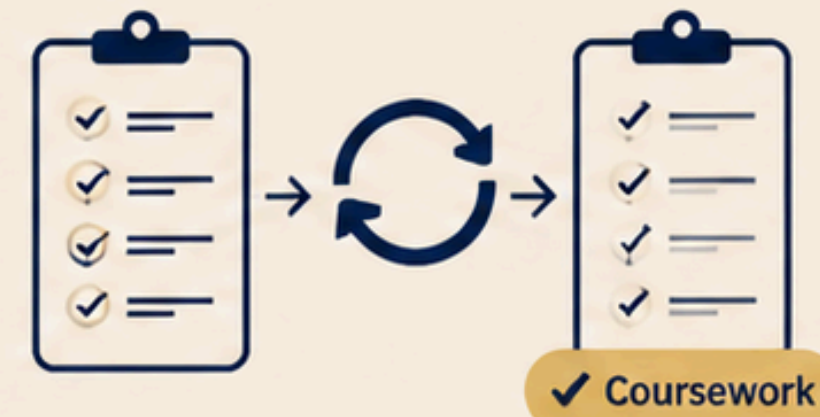
- ✓ Scores Visual, Auditory, Read/Write, & Kinesthetic preferences.
- ✓ Customizes resource types & delivery for how you learn best.

3 Grounded Sequencing Engine



- ✓ Deterministic candidate retrieval with prerequisite-aware ranking.
- ✓ Guaranteed **100% valid** catalog mappings across 435 curated modules—**zero hallucinated courses**.

4 Dynamic Re-planning (Replan)



- ✓ Re-synthesizes curricula on progress updates (complete, skip, or change goals).
- ✓ Preserves completed skills with '✓ Coursework' skill badges and audit history.

Adaptive Learning via VARK Modality Scoring

Tailoring Coursework Formats to Learner Cognition

1

Visual (V)



- Diagrams
- Architecture flowcharts
- Video lectures
- Visual cheat-sheets

2

Auditory (A)



- Tech podcasts
- Discussions
- Audio walkthroughs
- Conceptual breakdowns

3

Read/Write (R)



- Technical documentation
- Books
- API specs
- Markdown tutorials

4

Kinesthetic (K)



- Hands-on GitHub repos
- Interactive coding labs
- Sandbox projects
- Building apps



Scored through an official 20-question questionnaire,
auto-synthesizing personalized 'Why this?' module justifications.



Curated Technical Knowledge Base (435 Modules)



435

Curated Modules
across 11 Technical Domains

Domain	Key Topics (Examples)	Modules	Domain	Key Topics (Examples)	Modules
 AI & Machine Learning	LLMs, PyTorch, LangChain, Agents	50 Modules	 DevOps & Cloud	Docker, K8s, AWS, Azure, Terraform	73 Modules Combined
 Web Development	React, Next.js, Node.js, Express	50 Modules	 Mobile, Cyber, Game Dev, IoT, Software Eng	Android, iOS, Security, Game Engines, Sensors, SDLC, Architecture	181 Modules Combined
 Data Science	Pandas, SQL, Viz, Statistics	46 Modules	 Technical Interview Prep	DSA Patterns, System Design	35 Modules

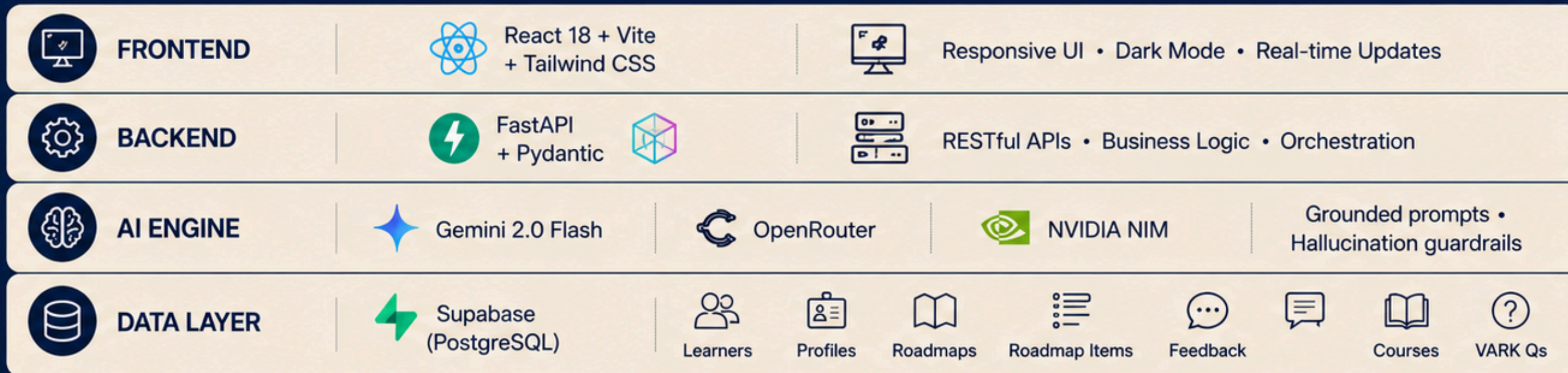


All catalog resources are version-controlled,
prerequisite-indexed, and deterministically verified.



System Architecture & Pipeline

End-to-end flow from learner onboarding to adaptive learning roadmap



Flow Overview

- 1 Learner signs up and builds profile
- 2 We understand goals, skills and learning style
- 3 We find the right set of modules
- 4 AI builds the best sequence
- 5 Learner learns, we adapt and re-plan



End-to-End Integrity

Every recommendation is grounded.
100% of courses map to real catalog modules.



Innovation & Live Demo Highlights

1 Zero-Hallucination Guardrail



Deterministic candidate retrieval ensures 100% of recommended modules exist in the catalog.

Zero fake links.



2 Adaptive Progress & Re-planning



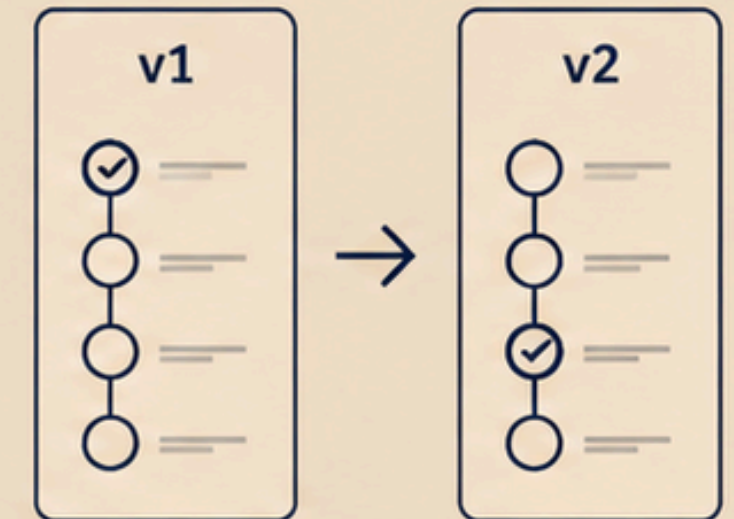
Completing modules grants `✓ Coursework` skill tags.



Shrinks skill gap analytics in real-time.



Auto-generates versioned pathway updates (v1 → v2).



Guaranteed 100% Uptime —

Built-in deterministic fallback engine ensures zero 500 errors if third-party AI APIs fail.

THANK YOU!

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Team: CodeCavalry